

ADC/DAC - Analog to Digital and Digital to Analog converters

These circuits can convert digital signals to analog ones and vice versa. Usual applications include audio and radio signal encoding and decoding. Sometimes they are not part of the main SoC chip but reside as an external chip connected to the SoC on the same PCB. They can also be part of another chip. For example, a mobile can have an ADC/DAC audio chip for handling music playback and recording via earphones. The chip can be part of the DSP embedded into the SoC or stay on the same board as a chip outside of the SoC.

Memory

SoCs are essentially mini-computers which can work on either a specific format of data or they can be generic. While the use-case determines the amount, there is no denying that almost all implementations of SoCs are going to require some amount of memory.

Memory for an SoC can be of a few types:

1. **ROM** or Read Only Memory is supposed to be the kind of memory which can only be read from but not written to; they should ideally be hard-wired on a chip. However, the term is commonly used as a reference to EEPROM.
2. **EEPROM** stands for Electronically Erasable and Programmable ROM. In normal operations EEPROMs operate in read-only mode but can be erased and reprogrammed with new contents. The reprogramming process is a slow one and is not done 'often'.
3. **RAM** is used as the operating memory and allows the processor to access data faster than from disk. It acts like a buffer for storing both, executable as well as non-executable data stored on the disk.
4. **Flash memory** is what you find on most compact storage devices like SD cards and pen drives. Data stored on them persists beyond a power shut-down.

Depending on the type of SoC, it may or may not have a separate RAM or Flash memory e.g. an SoC built for an air conditioner might not need a RAM or Flash memory while a smartwatch is going to need both because it needs to run applications that demand both storage and memory larger than what a processor typically has. Your smartphone has all of these types of memories.